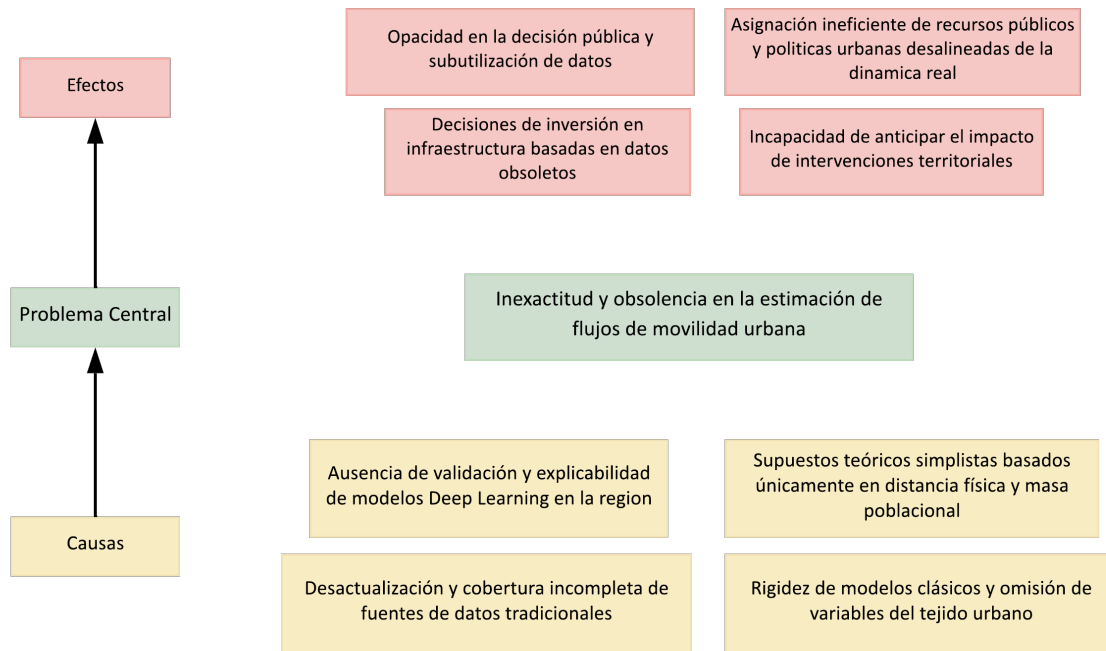






$$n\mathcal{Z}=\{z_1,z_2,\ldots,z_n\}T_{ij}z_iz_j$$

$$\in \mathbb{R}^{n \times n}, T_{ij} \geq 0 \forall i, j \in \{1, \ldots, n\}$$

$$ij$$

$$\mathcal{Z}O_i=\sum_jT_{ij}\hat{T}_{ij}(i,j)$$

$$T_{ij}=k\cdot \frac{P_i^{\alpha}\cdot P_j^{\beta}}{d_{ij}^{\gamma}}$$

$$P_iP_jd_{ij}k\alpha\beta\gamma$$

$$\hat{T}_{ij} = O_i \cdot \frac{P_j^\beta f(d_{ij})}{\sum_k P_k^\beta f(d_{ik})}$$

$$O_i = \sum_j T_{ij} i f(d_{ij}) e^{\beta_2 d} d^{\beta_2}$$

$$O_iD_j=\sum_iT_{ij}$$

$$y\Delta sxs$$

$$\Delta y = a \cdot \frac{\Delta x}{x}$$

$$s$$

$$y = a \cdot f(s) + c$$

$$f(s)c$$

$$m_l m_l$$

$$s_{ij}id_{ij}ijS_{ij}=m_i+s_{ij}+m_j$$

$$zp(z)$$

$$i\alpha ij$$

$$P_{ij} = \frac{e^{-\alpha(S_{ij}-m_j)} - e^{-\alpha S_{ij}}}{1 - e^{-\alpha M}}$$

$$m_jjS_{ij}d_{ij}i\alpha M=\sum_k m_kjm_js_{ij}$$

$$\alpha$$

$$zp(z)ijjd_{ij}iij$$

$$p(n_j \mid m_i, s_{ij}) = \frac{m_i \, n_j}{(m_i + s_{ij})(m_i + n_j + s_{ij})}$$

$$m_iin_jjs_{ij}d_{ij}iijij$$

$$\langle T_{ij} \rangle = T_i \cdot \frac{m_i \, n_j}{(m_i + s_{ij})(m_i + n_j + s_{ij})}$$

$$T_i = \sum_j T_{ij}i\sum_j \langle T_{ij} \rangle = T_i$$

$$L\in\mathbb{R}^d$$

$$^{(0)}=\sigma\big(W^{(0)}\big)\,,\,^{(\ell)}=\sigma\big(W^{(\ell)\,(\ell-1)}\big)\,,\ell=1,\dots,L-1$$

$$W^{(\ell)}\ell\sigma(\cdot)(0,x)(\alpha x,x)\alpha$$

$$ij$$

$$p_{ij}=\frac{(\boldsymbol{\beta}^\top(l_i,l_j))}{\sum_k(\boldsymbol{\beta}^\top(l_i,l_k))}$$

$$(l_i,l_j)=[P_j,\,d_{ij}](i,j)\boldsymbol{\beta}=[\beta_1,\beta_2]$$

$$H=-\sum_i\sum_j\frac{T_{ij}}{O_i}\,p_{ij}$$

$$\boldsymbol{\beta}^\top(l_i,l_j)f_\theta((l_i,l_j))i$$

$$p_{ij}=\frac{(f_\theta((l_i,l_j)))}{\sum_k(f_\theta((l_i,l_k)))}$$

$$ij\hat{T}_{ij}=O_i\cdot p_{ij}$$

$$f_\theta$$

$$^{(0)}=\big(W^{(0)}(l_i,l_j)\big)$$

$$^{(\ell)}=\big(W^{(\ell)\,(\ell-1)}\big)\,,\ell=1,\dots,14$$

$$f_\theta((l_i,l_j))=\top^{(14)}$$

$$=0,9=5\times10^{-6}$$

$$(l_i,l_j)\in\mathbb{R}^{39}$$

$$(l_i,l_j)=[(l_i),\,(l_j),\,P_i,\,P_j,\,d_{ij}]$$

$$(l)\in\mathbb{R}^{18}lP_iP_jd_{ij}$$

$$(\hat{\cdot},\cdot)=\frac{2\sum\limits_{i,j}\left(\hat{T}_{ij},\,T_{ij}\right)}{\sum\limits_{i,j}\hat{T}_{ij}+\sum\limits_{i,j}T_{ij}}$$

$$[0,1]$$

$$^i\phi_i(f,)=\sum_{S\subseteq\mathcal{F}\backslash\{i\}}\frac{|S|!\left(|\mathcal{F}|-|S|-1\right)!}{|\mathcal{F}|!}\left[f(S\cup\{i\})-f(S)\right]$$

$$\mathcal{F}Sif(S)S\phi_i i$$

$$f())'i$$

$$_i()=(x_i-x'_i)\cdot\int_0^1\frac{\partial f('+\alpha(-'))}{\partial x_i}\alpha$$

" =

amenity=hospitallanduse=residential

$$\lambda$$

$$jij d_{ij}$$

$$P_{ij} = \frac{m_j}{S_{ji}}$$

$$m_jjS_{ji} = m_j + s_{ji} + m_id_{ij}j$$

$$ij$$

$$P_{ij} = \frac{m_j}{S_{ij}} = \frac{m_j}{m_i + s_{ij} + m_j}$$

$$S_{ij} = m_i + s_{ij} + m_j d_{ij} i$$

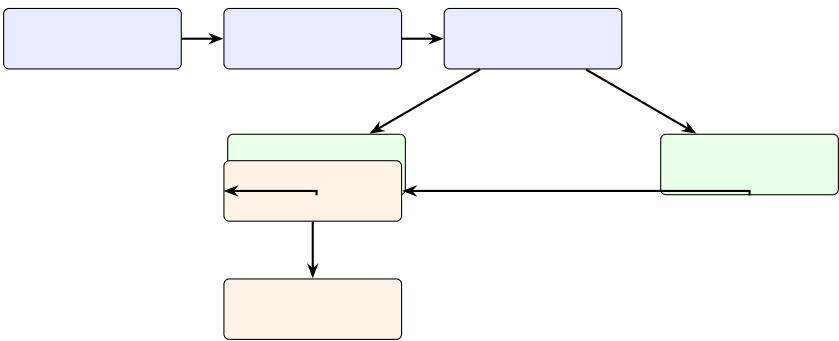
$$ij$$

$$P_{ij} = \frac{(m_i + \alpha \, s_{ij}) \, m_j}{\left[m_i + (\alpha + \beta) \, s_{ij}\right] \left[m_i + (\alpha + \beta) \, s_{ij} + m_j\right]}$$

$$\alpha \geq 0 \beta \geq 0 \alpha + \beta \leq 1$$

$$\alpha = 0 \beta = 1$$

$$\alpha = 1 \beta = 0$$



ij

```
viajes.(id_tarjeta, id_viaje)etapas.(id_etapa, correlativo_viajes)etapas
viajes

factor_expansionviajes
```

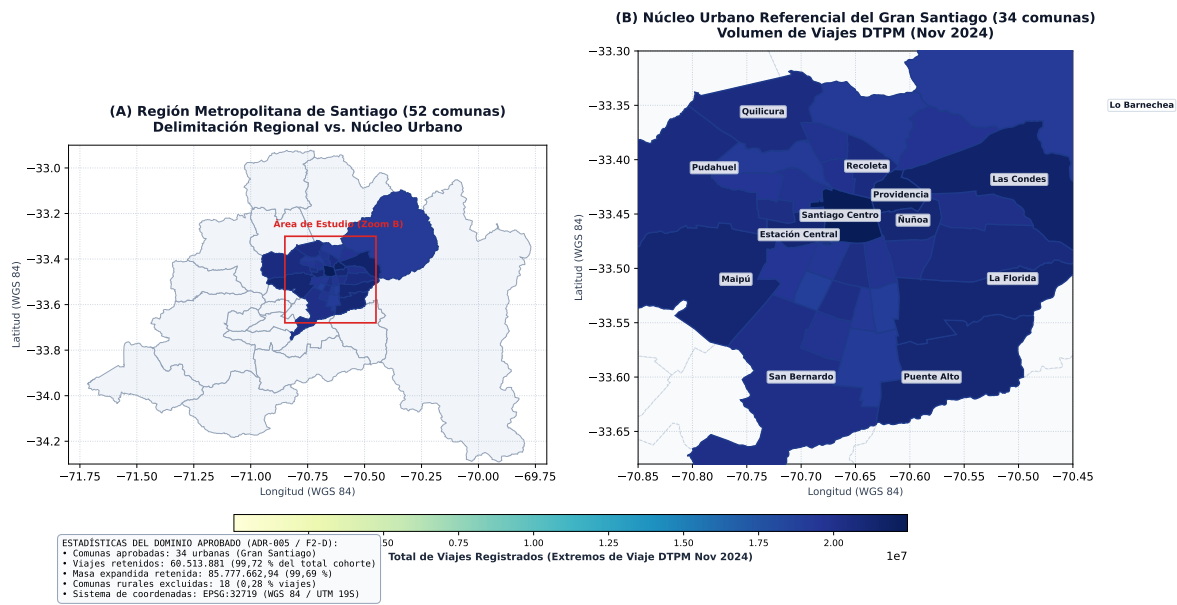
```
viajes      factor_expansion
etapas
ShapeZona
```

paper19

$$T_{ij} = \sum_{v \in \mathcal{V}_{ij}} w_v,$$

```
 $\mathcal{V}_{ij}$ ijwvtrip_countexpanded_mass
```

contains

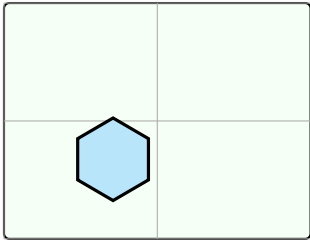
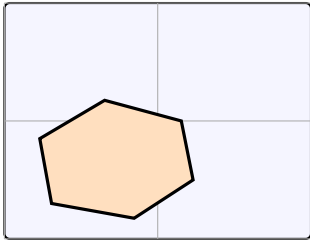


always_xy=True
latlng_to_cell

$$(x/15000)(y/15000)$$

paper19macro12_geometry20

$$_{ij} = [^{(19)}_i, ^{(19)}_j, d_{ij}],$$



$$_{ij}d_{ij}$$

$$\log 1p$$

$$V_{ij}$$

$$p_{ij}=\frac{(V_{ij})}{\sum_{k\in\mathcal{D}_i}(V_{ik})},$$

$$\mathcal{D}_i iV_{ij}=\beta_1\left(P_j\right)+\beta_2\left(d_{ij}\right)P_j^{\beta_1}d_{ij}^{\beta_2}V_{ij}=\alpha\left(P_j\right)-\beta d_{ij}$$

$$f_{\theta}(i_j)$$

$$\text{destinations_per_originpositive_destination_fraction}$$

$$\mathcal{L} = - \sum_i \sum_{j \in \mathcal{D}_i} y_{ij} \; p_{ij}.$$

$$5\times10^{-6}$$

$$=\frac{2\sum_{ij}(T_{ij},\widehat{T_{ij}})}{\sum_{ij}T_{ij}+\sum_{ij}\widehat{T_{ij}}}.$$

<https://www.dtpm.cl/index.php/sistema-de-transporte-publico-de-santiago>

